

Practice Quiz 07C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

State the epsilon-delta definition and use it once.

1. (a) State, with all quantifiers, the definition of $\lim_{x \rightarrow c} f(x) = L$.

Solution

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x (0 < |x - c| < \delta \rightarrow |f(x) - L| < \varepsilon). \quad \blacksquare$$

- (b) For $f(x) = 5x + 1$, prove from the definition that $\lim_{x \rightarrow -1} f(x) = -4$.

Solution Let $\varepsilon > 0$ and choose $\delta = \varepsilon/5 > 0$. Let x be arbitrary and suppose $0 < |x + 1| < \delta$. Then

$$|f(x) - (-4)| = |5x + 5| = 5|x + 1| < 5\delta = \varepsilon.$$

This verifies the definition, so $\lim_{x \rightarrow -1} (5x + 1) = -4$. $\blacksquare$